Hail Impact Resistance of Building Materials Testing, Evaluation and Classification

Peter Flüeler [1]
Maja Stucki[2]
Fabio Guastala[3]
Thomas Egli [4]

T 32

ABSTRACT

Results are presented from an extensive research project on the hail impact resistance of various building materials and components including tiles, metal sheets, skylights, wood, glass, flexible sheets, exernal thermal insulation and finished systems, window shutters. The data established by impacting specimens with simulated hailstones made of plastic balls and ice spheres are correlated to each other. The results are discussed and recommendations are given for a new testing practice which will be implemented in Switzerland. In the project, it became evident that high speed hail tests can not be replaced by a dart impact test using falling steel balls to achieve the same kinetic energy. The values of hail impact resistance (HIR) show a marked difference between high modulus and low modulus materials, and their structural stiffness and surface properties.

A new classification system for the hail impact resistance of building materials is proposed. This system considers the five predominant damage categories experienced during impact; specifically, loss of functionality, deformation, fracture, aesthetics, and damage affecting aging. The results also show that requirements of certain standards such as that for solar modules do not comply with the hail size observations made during naturally-occurring hail storms.

KEYWORDS

Hail impact resistance, Testing, Correlation ice spheres vs. plastic ball impact, Classification

[1] Swiss Federal Laboratories for Materials Testing and Research EMPA, dept. civil and mechanical engineering, CH-8600 Dübendorf Switzerland, Phone +41 44 823 55 11, Fax 823 4026, peter.flueler@empa.ch and
FPC Flüeler Polymer Consulting GmbH, CH-8607 Aathal Switzerland, p.flueler@hispeed.ch

[2] Swiss Federal Laboratories for Materials Testing and Research EMPA, dept. civil and mechanical engineering, CH-8600 Dübendorf, Switzerland, internship at EMPA as trainee

[3] Egli Naturgefahren, CH-3001 Bern Switzerland Phone +41 31381 52 90, Fax+41 31 381 5291, stucki@naturgefahr.ch

[4] Egli Naturgefahren, CH-9001 St. Gallen Switzerland, Phone +41 71 274 7148 , Fax +41 71 274 7149, egli@naturgefahr.ch

1 INTRODUCTION

In Switzerland, each year hailstorms cause substantial damage to building envelopes and over time, total losses show an upward trend. The use of a plastic ball to simulate hail impact started in Switzerland in 1970 when roofing membranes were competing against traditional roofing materials like roof tiles. Standards such as SN 564 280 [1], recommendations of building authorities and insurance companies referred to such procedures. In determining hail impact resistance (HIR), a 40 mm plastic ball is shot at a specified speed at the test specimen. Temperature sensitive materials are cooled at the surface to 5° C and positioned on a rigid and/or flexible support. Facade elements are fastened at an angle of 45° or 90° with regular fasteners, and jointed to each other. The resultant damage is assessed for leakage and /or visual deficiencies.
Constant projectile properties, reproducibility, instant damage assessment, time and costs savings are the advantages of the current test procedures. There are, however, various disadvantages in this procedure in regard to the natural weather influences. In general, the density of plastic balls is higher than that of natural hailstones and the fracture behaviour is brittle for ice, elasto-plastic for plastic balls. This means that a plastic ball exerts higher energy than that occurring with a natural hail stone.

2 HIGH MASS - LOW SPEED VERSUS LOW MASS - HIGH SPEED

The approach to this problem with reference to natural environmental condition demands knowledge of the impact speed of a hailstone. It can be calculated from analytical evaluation or – as in recent times – from the use of measuring devices during a hail storm. If a high speed camera is available, the impact speed can be indirectly established by video-recording the impact.

2.1 Evaluation of Impact Velocity

At impact on the ground, hail appears to be of white colour. This fact leads to the conclusion that the amount of trapped air in the ice is apparently rather high. With increasing size, however, sliced hailstones often show a shell-like structure. Within this structure, clear ice alternates with porous ice resulting from many ups and downs in the turbulent air circulation within a storm super cell. Size of hailstone and aerodynamic drag c_w as determined by the hailstone shape has the strongest influence on impact velocity.

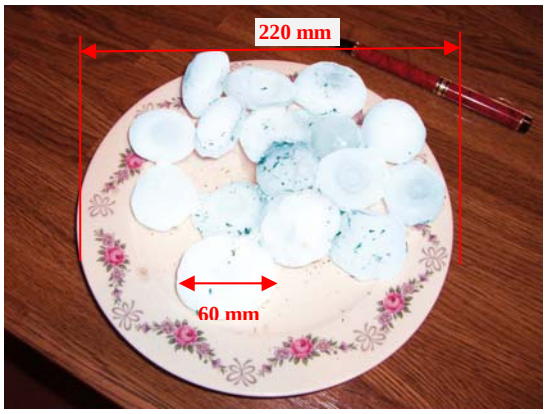

Figure 1. Shapes and dimensions of collected hailstones that fell from a super cell near Lugano on June 21, 2007 (photo by F. Terrasi)

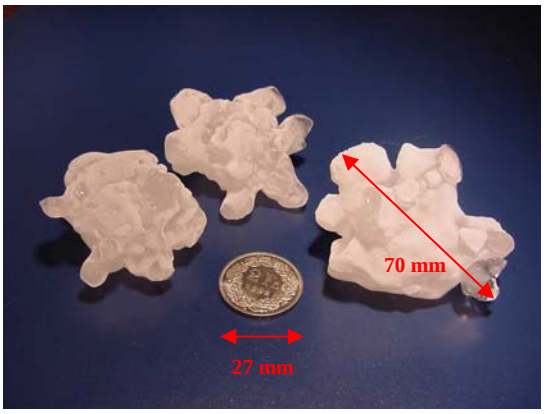

Figure 2. Shapes and dimensions of collected hailstones that fell from a super cell in the city area of Zurich on June 24, 2002 (photo by U. Spreiter; the diameter of the 2-Swiss Francs coin is 27 mm.)

These shapes vary from round, egg-shaped, disk-like, smooth, warty, bulgy, and even extremely jagged surfaced. Recently, bulgy, pointed forms were detected also in smaller-sized hailstones (Figure 1 and 2). Therefore the aerodynamic drag coefficient, c_w, can range from 0.45 for a smooth surface to 0.8 for a rough surface. In turn, the density, ρ_{ice}, may range from 0.60 to 0.91 kg/dm^3.

In the past, various scientists have reported on the hail impact velocity and its destructiveness. As an example: In 1937, Bilham and Relf [2] studied differences between small (< 10 mm) and large hailstones. Motz [3], Kawashita/Flüeler [4], and others evaluated the most cited equations for terminal velocity. Figure 3 shows the range of the calculated terminal velocity, v_{tH}, for hailstone diameters ranging from 20 to 100 mm. As a rule, factors in the calculation include the sphere diameter, d_H, the density of the ice, ρ_{ice}, the density of the air, ρ_{air}, and the coefficient of aerodynamic drag in air, c_w. The following formula is most frequently used:

$$v_{tH} = \sqrt[2]{\frac{4 \cdot d_H \cdot g \cdot \rho_{ice}}{3 \cdot c_w \cdot \rho_{air}}} \qquad (1)$$

$$E_{kin} = m \cdot v^2/2 \qquad (2)$$

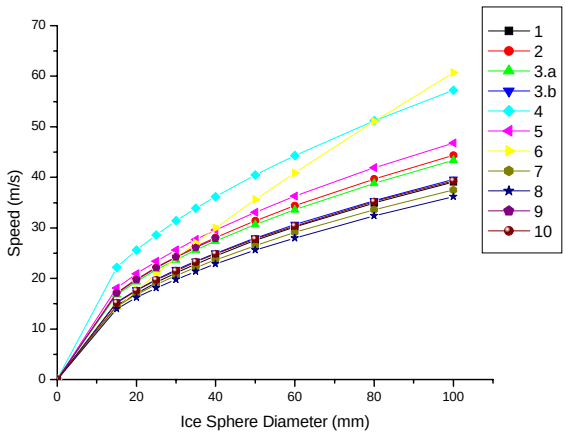

Figure 3. Calculated terminal velocity for round hailstones from different authors: 1. Bohm, 2. ASTM 1038-05, 3a. EMPA with c_w 0.5, 3b. EMPA c_w 0.6, 4. Motz ρ_{air} 0.9,kg/m^3, 5. Motz ρ_{air} 1.23,kg/m^3, 6. Heymsfield, 7. Pflaum, 8. and 9. Matson, 10. Lozowski, Guastala and Flüeler

In this present study, the following constants are used (Figure 2, curve 3a): Ice density, ρ_{ice} : 875 kg/m^3, air density at 20 ° C, ρ_{air} : 1.226 kg/m^3, and coefficient of aerodynamic drag, c_w: 0.5 kg/m^2.

2.2 Test Practices of Common Standards

Since the beginning of recorded history, Australia, South America, the United States, and Central Europe have known the effects of hailstorms. To toughen materials against hail impact, test procedures were first established in South Africa in the 1950s and in the United States at NIST in the 1960s. In Switzerland, the plastics industry provided evidence in the early 1970s that polymer roofing membranes are equal or even superior to classic roofing materials such as clay tiles. Initial impact testing was conducted using ice spheres. Nevertheless, for practical and economical reasons, a 40 mm polyamide (PA) sphere was chosen. Comparisons of in-service hailstone damage on polymer roofing membranes to that obtained with the PA spheres showed similar damage patterns. Due to this fact, SIA established a test protocol in 1977 [1] with the requirement of a velocity of 17 m/s when shooting at a chilled waterproofing membrane supported by a steel plate and soft thermal insulation board (EPS).

This velocity equates to a kinetic energy of 5.6 J. In comparison, good classic clay tiles become damaged at a velocity of 9 m/s which is equal to 1.6 J.

3 TEST PROGRAM

To simulate the conditions of natural hail and its impact on the building envelope, an extensive experimental investigation was performed. A comparative study was carried out on 50 types of materials originating from 11 fields of application using laboratory-made ice spheres and plastic balls.

3.1 Test Apparatus and Parameters

The test apparatus consisted of a pneumatic gun positioned vertically (Figure 4). This apparatus was originally designed for use with plastic balls only. In the course of expanding our technical investigations, the apparatus was modified to launch 15 to 50 mm ice spheres. By using a light beam, the projectile velocity is instantaneously measured at the end of the gun barrel in a distance of 30 cm. Target and distance are exactly aimed at by two focussing lasers (Figure 5). The velocity is set by the pneumatic pressure corresponding to calibration values enabling a repeatability of < 1 m/s.

Figure 4. Test apparatus and tilted steel frame with mounting for a wood framed window glass

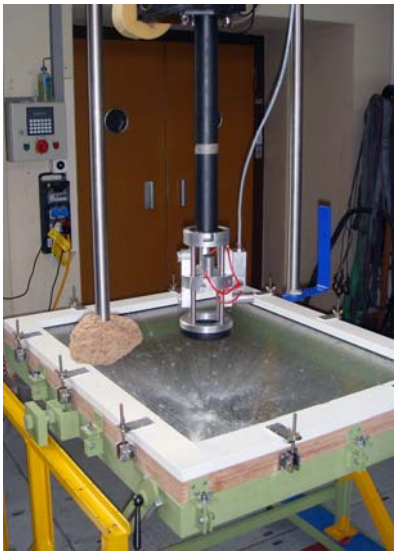

Figure 5. Edge impact of a 50 mm ice sphere on a 6 mm thick wire glass window

For the project, the following parameters were chosen:

Set-up of test specimen:	Condition new, as installed in building, custom-designed
Size of test specimen:	0.8 to 1.1 m^2 with jointing, possible overlaps, original fasteners
Surface treatment:	Temperature sensitive materials: chilled with ice granules for 3 min.
Test temperature:	Room temperature and humidity, approx 23° C/50 % r. h.
Impact angle:	Roofing 90°, façade 45°, both applications 45° and 90°
Type of impact:	Single shot mode
Impact speed:	Appropriate speed for size of projectile

3.2 Projectiles and Properties

For projectiles, precision balls made from PA 66 (density of 1.16 kg/dm^3) and laboratory ice spheres (density 0.875 g/cm^3) were used. The ice spheres were made in silicon moulds using demineralised water. The production of crack-free and essentially pore-free ice spheres required approx. 17 hours at -20° C in a freezer. Freezing of the water was induced by a freezer plate at the bottom of the mould, thus facilitating slow growth of the ice core from bottom to top. Due to time-dependent changes of ice,

a shelf time of less than 3 weeks was observed for the newly made ice spheres. The important properties of both types of projectiles are listed in Table 1:

Table 1. Mass of ice spheres and plastic balls for various diameters.

Type of sphere	Unit	Diameter (mm)				Failure behaviour
		20	30	40	50	
Mass of ice	(g)	3.8	12.3	30.2	58.3	brittle
Mass of Polyamide	(g)	4.8	16.1	38.8	74.9	tough

3.3 Procedure

To investigate the damage mechanism and weak points, test specimens were first impacted using PA-balls fired within the velocity range observed for real hailstones at impact. At a damage velocity considerably higher or lower than natural velocities, test specimens were impacted by sphere sizes 10 mm higher or lower. Then, the procedure was applied by using ice spheres.

4 DAMAGE ASSESSMENT AND EVALUATION

Assessment of damage in regard to insurance codes of practice was the most challenging task. The diversity of materials, supports, substrates, and fastenings along with the wide-ranging damage characteristics at impact demanded very close examination of each application. Therefore, the damage characteristics due to impact were grouped within the following general categories:

a) Loss of a main function: such as water tightness, breakdown of mechanical/electrical properties, loss of load bearing property etc,
b) Deformation: indentation, dent formation, deformation of defined depth,
c) Cracking and fracture: visible cracks, spontaneous fracture, delamination,
d) Aesthetics: change of appearance, loss of light transmission, change of aesthetical appearance, view in back light from a distance of 5 m, and
e) Damage affecting aging: inherent cracking, face separation, de-bonding, damage of surface layers.

After impact, the damage category was assigned based on the lowest velocity (i.e., kinetic energy according to Eq. 2) that resulted in damage. For ductile materials such as metallic sheets, the formation of a dent was judged to be aesthetic damage. In general, aesthetic damage was not observable until the depth of the indent reached about 0.5 mm.

5 RESULTS

The diversity of the investigated materials and their uses generated a wide range of results. Nevertheless, these results could be classified into three material categories according to load bearing behaviour, i.e. stiffness (Tables 2.1 – 2.3). A condensed report is provided by [6], also in French.

Figure 6 is a plot relating depth of indentation to projectile kinetic energy for the case of a 0.7 mm zinc-plated, corrugated steel sheet impacted using 40 mm PA balls and 40 mm ice spheres. Figure 7 shows a photo of the indentations at projectile velocities of 26, 30, and 36 m/s. At a 90° impact angle, both projectile types caused circular indentations that increased in size with increasing velocity. At an angle of 45° mainly ellipsoidal indents resulted. Upon impact, PA balls remained intact and showed no cracking. In contrast, ice spheres tended to split - even at a velocity as low as 10 m/s - depending on the nature, mass and surface topology of the specimen. A clear observation from the testing was that, for heavy mass specimens such as clay tiles, the ice fragmentation pattern was distinct and more diverse changing with increasing velocity in comparison to the fragmentation of compliant specimens. A heavy mass specimen is one for which the ratio of the test specimen mass to the projectile mass is

greater than 50. Another observation was that low HIR values (i.e., < 3 J) were in general found for specimens where the damage was categorized as aesthetic rather than functional.

Table 2.1. High stiffness, tough materials

Material/Component	Element stiffness	E·I (kN·mm²/mm)	Angle (°)	E_{kin} (J)	HIR-class	Lowest values for Product	Ø (mm)	E_{kin} (J)
Tiles	high	2330-2635	90	13.7 - 27.3	4	clay tiles	50	13.75
Glass	high	370 - 3000	45	5.5 - 46.7	3 - 5	wire glass 7	40	5.5
Fibre cement	high	140-252	45/90	17.6 - 38.2	4 - 5	corrug. plate 5.5	40	17.6
Polymer plates	high	0.3-9.6	90	6.4 - 38.8	3 - 5	PMMA 4	30	6.4
Skylights	high	5.3 - 7.4	90	0.8 – 3.1	2 - 3	PMMA 2.5	30	0.8
GRP boards	high	6-88	45	1.4 - 20.5	2 - 4	GRP-UP struct.	30	1.4

Table 2.2. Low stiffness materials, non load bearing

Material/Component	Element stiffness	E·I (kN·mm²/mm)	Angle (°)	E_{kin} (J)	HIR-class	Lowest values for Product	Ø (mm)	E_{kin} (J)
Shutters	very low	0.1-4.2	45/90	0.05 – 1.75	1 - 2	profile, foam 0.25	20	0.05
Roller blinds	very low	0.5-1.2	45/90	0.2 – 0.7	1 - 2	folded 0.45	20	0.2
GRP: corrug., trapez.	low	1.95	90	0.4 – 0.5	1	GRP-UP trp. 1.4	20	0.4
Metal sheets, facade	low	1.9-2.6	45	0.6 – 1.7	1 - 2	alu 0.7	30	0.6
Membrane: stiff*	medium	0.001-0.6	90	39 - 90	5	SBS 3.7 sand coat	50	> 80
Membrane: soft*	medium	0.001-0.6	90	12.9 - 53.3	4 - 5	SBS 3.7 sand coat	40	12.9

Table 2.3. Medium stiffness materials

Material/Component	Stiffness	E·I (kN·mm²/mm)	Angle (°)	E_{kin} (J)	HIR-class	Lowest values for Product	Ø (mm)	E_{kin} (J)
Larch wood	medium	1.7	45	0.6 - 1.8	1 - 2	coated	30	0.6
Spruce wood	medium	1.5	45	0.8 - 3.1	2	planed	30	0.8
EIFS**	medium	1.7 – 2.8	45	5.7 - 17.0	3 - 4	EPS 20, 4	30	5.7
Metal sheets, roof	medium	2.1 - 6.0	90	0.6 - 2.0	1 - 2	cupper, 0.6	30	0.6

*membrane placed on stiff / soft substrates, respectively **External Insulation and Finished System

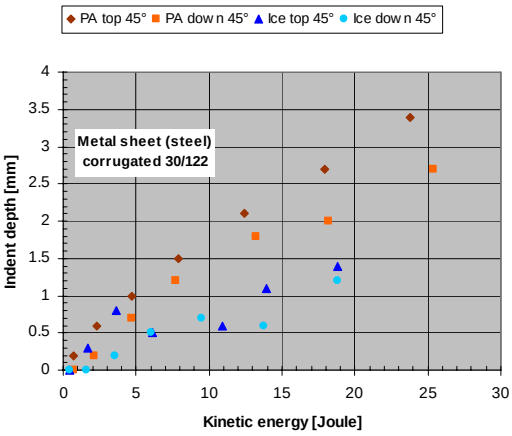

Figure 6: Indentation versus kinetic energy (J) of a corrugated 0.7 mm steel sheet tested using 40 mm PA balls and 40 mm ice spheres

Figure 7: Indentations caused by 40 mm projectiles; impact using PA balls (left) and ice spheres (right) at velocities of 26, 30 and 36 m/s

6 CORRELATION OF DATA FROM POLYAMID AND ICE SPHERES

Test data from two projectile materials - ice versus Polyamide - allow a correlation of the tested materials. A ratio (I/P) can be calculated between the values of velocity (and also kinetic energy) of ice (I) spheres and Polyamide (P) balls sphere. Figure 8 shows the relationship between the primary kinetic energy data for various specimens tested using 40 mm PA balls and 40 mm ice spheres.

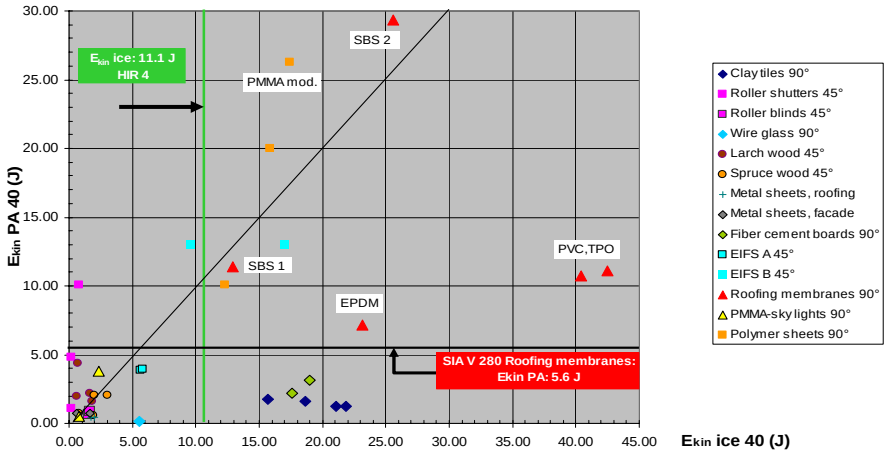

Figure 8. Kinetic energy in J for ice spheres and PA balls causing damage. Actual reqirement (red) for roofing membranes in Switzerland

This figure confirms clearly that many specimens experienced damage at very low kinetic energies. It is noted that hailstones having diameters less than 30 mm generally have kinetic energies of less than 3.5 J. Most important from Figure 8, it is evident that the data points for many specimens fell well below or above the correlation line indicating for these cases that testing needs to be conducted with ice spheres and not with PA balls. This is especially the case for roofing membranes tested on rigid substrates and for clay tiles.

7 CLASSIFICATION AND DESCRIPTION

A classification of building materials should be understood not only by material scientists and professionals in the building business but also by users. So, it should be ease to understand and related to the observed weather phenomena. In a 1991 publication, Flüeler [5] made an attempt to define a classification system including 5 levels of hail impact resistance. It is now stated to classify hail impact resistance into classes 1 to 5 (Table 3) that corresponds to hailstone diameters of 10 to 50 mm at which the building materials remain damage-free. It includes the corresponding terminal velocity of an impacting hailstone, as calculated using equation 1, curve 3a of the plot in figure 2 and calculated kinetic energy using equation 2. An examination of Table 3 shows that for each HIR class, the range of kinetic energy corresponding to that given HIR Class is rather large, for example, ranging from 11.1 J to < 27.0 J for HIR Class 4. This realization dictates that, for a more precise description differences in kinetic energy sustained without damage must be taken into account in the proposed classification system. The classification thus signifies not only the HIR class designation, but also the highest kinetic energy achieved without damage for the given sized ice sphere. Table 4 provides HIR Class designations (without kinetic energy) for the specimens in this study.

Table 3: Classification of hail impact resistance (HIR) and calculated kinetic energy

HIR-Class	Ice Sphere diameter (mm)	Mass (g)	Terminal velocity (m/s)	Kinetic energy (J)
1	10	0.5	13.8	***0.04***
2	20	3.6	19.5	***0.7***
3	30	12.3	23.9	***3.5***
4	40	29.2	27.5	***11.1***
5	50	56.9	30.8	***27.0***

Table 4: HIR-guide values of tested building materials and predominant damage category

HIR Class	Material/Component	Type	Thickness (mm)	Predominant Damage Category
1	Roller shutters, roller blinds	aluminium, folded	< 0.5	Deformation, aesthetics
	Metal sheets	cupper, tin	< 0.6	Deformation, aesthetics
	GRP-panels	light, shaped	> 2	Fibre matrix defect
2	Metal sheets façade and roofing	Fe, Cu-Fe, Ti plated	>0.5 - 0.7	Deformation, aesthetics
	Roller shutter reinforced	aluminium profile	0.9	Deformation, aesthetics
	Wood panels	planed, coated	25	Indent, cracked paint
3	Skylights	PMMA	2.5	Fracture, leakage, transparency
	GRP-double face panels	structural	20	
	EIFS (EPS and rock wool)	reinforced glass web	>3	Fibre matrix defect
	Wire glass	wire net 10 mm	7	Cracking, de-bonding Fracture
4	Roofing membranes	EPDM, SBS	4	Leakage
	Polymer sheets	PMMA modified	4	Transparency, defect
	Roof tiles	clay, structured surface	15	Internal fissure (sound check)
	Fibre cement boards/roofing	flat, undulated	6	Indent, surface crack, fissure
5	Roofing membranes	TPO, PVC-P	>1.5	Leakage
	Polymer sheets, skylights	PC, plain + structured	4	Deformation
	Safety glasses	single, laminated	6	Splitting, fracture
	Window glasses	insulated, alum. framed	4/16/4	Splitting, fracture

Note: HIR Classes in this table are examples achieved in the study and should not be taken as requirements or guide values.

8. CONCLUSIONS

1. This study confirmed field observations particularly from insurance companies that indicate that considerable hail damage payment (> 80 %) is made for the materials that fall within the HIR Class 1 and 2. They experience damage at kinetic energies of 0.7 J or less.

2. Impact of stiff, high-mass construction materials by a PA ball provokes considerably higher kinetic energy than that of an ice sphere of the same size. The fragmentation energy is not available for the damage process. For very stiff and high-mass materials, ice spheres have a kinetic energy up to 15 to 20 times greater than that of a PA ball to inflict damage.

3. With the exception of mass difference, lightweight elastic materials behave almost similarly with the two types of projectiles because of low fracture/deformation energy absorption.

4. For materials with structured cross sections with thin faces (e.g., double face plate, < 1 mm), a smaller projectile diameter might provoke damage while a larger diameter causes no damage.

5. Very often, hail damage resulting in loss of functionality or other physical characteristics is not as disconcerting as aesthetic damage. As an example, car bodies dented by hail are still roadworthy.

6. The effect of hail impact on the aging behaviour must be sufficiently taken into consideration.

7. Due to the diversity of materials and systems used for building envelopes, HIR evaluations have to be performed using ice spheres impacting at natural terminal velocity. Moreover, the kinetic enegy sustained during these evaluations has to be taken into account in an HIR classification system.

ACKNOWLEDGMENTS

The research was funded by the prevention fund of the Swiss association of cantonal building insurance companies. The authors wish to acknowledge the various contributions from the building material industries and the support of the EMPA laboratories, mainly polymer and composites.

REFERENCES

SIA, '*Polymer waterproofing membranes/Kunststoff-Dichtungsbahnen'*, Swiss Assoc. of Engineers and Architects, standard SN 584 280, editions 1977, 1988, 1996.

Bilham, E.G., Relf, E.F. 1937, '*The dynamics of large hailstones'*, Symon's Meteorogical Magazine, pp.149-162

Motz, H.D. 1986, '*Hagelrisiko- eine sicherheitswissenschaftliche Studie von Kausalität und Effekt'*, PhD-Thesis Univ. Gesamthochschule Wuppertal, D 468, Wuppertal, Germany.

Kawashita, L., Flüeler, P. 2002, Simulated Hail impact tests with ice and polyamid spheres, internal research report Empa no 860'091/2 (not public), August 20, 2002

Flüeler, P., Rupp, F. 1988, 'The *Hail Impact Resistance of Plastic Components of the Building Shell'*, Proceedings 3dbmc int. conference on durability of building materials, Bournemouth England.

Swiss Assoc. of Fire Insurance Companies VKF Bern, 7/2007, *Elementarschutzregister Hagel, Synthesebericht*, Untersuchungen zur Hagelgefahr und zum Widerstand der Gebäudehülle, 32 pages